

Fundamentals of Electrical Engineering: RL Circuit Transients

1. Introduction to Inductance and Inductors

Core Concepts & Intuition

An **inductor** is a passive electrical component that stores energy in a **magnetic field** created by an electric current passing through it.

Think of current flowing through an inductor like water flowing through a heavy water wheel:

- The water wheel resists any sudden change in the water's speed due to its inertia.
- Similarly, an inductor resists any **change in current** through it. It does not oppose the flow of steady current—it only opposes *changes* in current over time.

Fundamental Equations

1. Stored Energy (E):

The energy stored in an inductor's magnetic field depends on its inductance L and the square of the current I passing through it:

$$E = \frac{1}{2}LI^2$$

- E : Energy in Joules (J)
- L : Inductance in Henries (H)
- I : Current in Amperes (A)

2. Faraday's Law of Electromagnetic Induction & Induced EMF (e):

When magnetic flux ϕ changes through a coil of N turns, an electromotive force (EMF) is induced:

$$e = -N \frac{d\phi}{dt} = -L \frac{di}{dt}$$

- The negative sign represents **Lenz's Law**: the induced voltage opposes the change in current that created it.
- $\frac{di}{dt}$ represents the rate of change of current with respect to time.

3. Self-Inductance (L):

Inductance is defined as the flux linkage per unit current:

$$L = N \frac{d\phi}{di}$$

Types of Inductors

- **Air Core Inductors**: Possess a linear BH curve (relationship between magnetic flux density B and magnetic field strength H is a straight line). Inductance stays constant regardless of current.
- **Iron Core / Ferrite Core Inductors**: Possess a non-linear BH curve due to magnetic saturation effects of the core material.

2. Transient Behavior in Series RL Circuits

What is a "Transient"?

When a switch in an electrical circuit is thrown, the circuit transitions from an initial state to a new final state (**steady state**). The temporary, time-dependent behavior during this transition is called the **transient response**.

Because inductors oppose instantaneous changes in current, the current cannot instantly jump from 0 A to its final value. Instead, it grows gradually along an exponential curve.

 Diagram: Series RL Circuit with Switch

Visual Layout: A simple single-loop circuit containing a DC voltage source V on the left, a switch labeled $t = 0$ at the top, a series resistor R along the top rail, and an inductor L on the right side. The loop current is labeled as $i_L(t)$ moving clockwise. The voltage across the inductor is marked as $v_L(t)$ with positive polarity at the top and negative at the bottom.

Step-by-Step Derivation of Inductor Current $i_L(t)$

Applying Kirchhoff's Voltage Law (KVL) around the series loop after the switch closes at $t = 0$:

$$V = R \cdot i_L(t) + L \frac{di_L(t)}{dt}$$

This is a **first-order linear differential equation**. Its solution consists of two components:

$$i_L(t) = i_{c.f.}(t) + i_{p.i.}(t)$$

1. Complementary Function $i_{c.f.}(t)$ (Transient State):

Solves the homogeneous equation $Ri_L + L \frac{di_L}{dt} = 0$:

$$i_{c.f.}(t) = Ce^{-t/\tau} = Ce^{-\frac{R}{L}t}$$

where $\tau = \frac{L}{R}$ is defined as the **Time Constant**.

2. Particular Integral $i_{p.i.}(t)$ (Steady State):

At steady state ($t \rightarrow \infty$), current becomes constant ($\frac{di_L}{dt} = 0$), meaning the inductor acts like a short circuit:

$$i_{p.i.}(t) = \frac{V}{R}$$

3. Combining and Applying Boundary Conditions:

$$i_L(t) = Ce^{-\frac{R}{L}t} + \frac{V}{R}$$

At $t = 0^-$, the inductor current was zero ($i_L(0^-) = 0$). Since current through an inductor cannot change instantaneously:

$$i_L(0^+) = i_L(0^-) = 0$$

Substituting $t = 0$ into the equation:

$$0 = Ce^0 + \frac{V}{R} \implies C = -\frac{V}{R}$$

4. Final Closed-Form Equations:

• Inductor Current Growth Equation:

$$i_L(t) = \frac{V}{R} \left(1 - e^{-\frac{R}{L}t}\right)$$

• Inductor Voltage Equation:

$$v_L(t) = L \frac{di_L}{dt} = Ve^{-\frac{R}{L}t}$$

Key Properties & The Time Constant (τ)

The **time constant** (τ) dictates how fast the transient process occurs:

$$\tau = \frac{L}{R} \quad [\text{seconds}]$$

 Diagram: Exponential Current Growth Curve

Visual Layout: A graph showing time t on the horizontal axis and current $i_L(t)$ on the vertical axis. Starting at $(0, 0)$, an exponential curve rises smoothly toward a horizontal asymptote at $i_L = \frac{V}{R}$. A dashed tangent line drawn from $t = 0$ reaches $\frac{V}{R}$ precisely at $t = \tau$.

- At $t = 1\tau$: $i_L(\tau) = \frac{V}{R}(1 - e^{-1}) \approx 0.632\frac{V}{R}$ (63.2% of steady-state value).
 - At $t = 5\tau$: $i_L(5\tau) = \frac{V}{R}(1 - e^{-5}) \approx 0.993\frac{V}{R}$ (99.3% of steady-state value).
 - **Engineering Rule:** In practical designs, a duration of 5τ is universally assumed to be the time required for the circuit to reach steady state.
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3. Worked Examples

Worked Example 1: Basic Series RL Response

Problem Statement:

A circuit having a resistance of $20\ \Omega$ and an inductance of $0.5\ \text{H}$ is connected across a $200\ \text{V}$ DC supply. Calculate:

1. Equation of current as a function of time
 2. Time constant of the circuit
 3. Current at $t = 0.025\ \text{s}$
 4. Current at $t = 0.05\ \text{s}$
 5. Energy stored in the magnetic field when current reaches $5\ \text{A}$
 6. Rate of increase of current at the instant of closing the switch ($t = 0$)
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Solution:

Step 1: Identify Given Parameters

- $V = 200\ \text{V}$
- $R = 20\ \Omega$
- $L = 0.5\ \text{H}$

Step 2: Find the Time Constant (τ)

$$\tau = \frac{L}{R} = \frac{0.5}{20} = 0.025\ \text{s}$$

Step 3: Write the Current Equation $i_L(t)$

$$\frac{V}{R} = \frac{200}{20} = 10\ \text{A}$$

$$\frac{R}{L} = \frac{1}{\tau} = \frac{1}{0.025} = 40\ \text{s}^{-1}$$

$$i_L(t) = 10(1 - e^{-40t})\ \text{A}$$

Step 4: Calculate Current at $t = 0.025\ \text{s}$ ($t = \tau$)

$$i_L(0.025) = 10(1 - e^{-40(0.025)}) = 10(1 - e^{-1}) = 10(1 - 0.36787) = 6.32\ \text{A}$$

Step 5: Calculate Current at $t = 0.05\ \text{s}$ ($t = 2\tau$)

$$i_L(0.05) = 10(1 - e^{-40(0.05)}) = 10(1 - e^{-2}) = 10(1 - 0.1353) = 8.65\ \text{A}$$

Step 6: Compute Stored Energy at $I = 5\ \text{A}$

$$E = \frac{1}{2}LI^2 = \frac{1}{2}(0.5)(5)^2 = 0.25 \times 25 = 6.25 \text{ Joules}$$

Step 7: Compute Rate of Increase of Current at $t = 0$

From KVL at $t = 0^+$, since $i_L(0^+) = 0$:

$$V = R(0) + L \left. \frac{di}{dt} \right|_{t=0} \implies \left. \frac{di}{dt} \right|_{t=0} = \frac{V}{L} = \frac{200}{0.5} = 400 \text{ A/s}$$

Worked Example 2: Two-Interval Switching Sequence

Problem Statement:

In the network shown below, the switch is closed to Position 1 at $t = 0$, and then moved to Position 2 at $t = 1.4 \text{ ms}$.

1. Obtain the equation for inductor current for both time intervals.
2. Determine the current in the inductor at $t = 2.5 \text{ ms}$.

Diagram: Two-Position Switch RL Circuit

Visual Layout: On the left is a 10V DC source in series with a 5Ω resistor leading to terminal 1 of a single-pole double-throw switch. Terminal 2 connects directly to a parallel path with a 2.5Ω resistor returning to the source bottom rail. The switch wiper connects down through a 10 mH inductor in series with a 2.5Ω resistor to the ground rail.

Solution:

Interval 1: $0 \leq t < 1.4 \text{ ms}$ (Switch at Position 1)

Step 1: Simplify the Circuit Parameters

- Total resistance in series with inductor: $R_{eq1} = 5 \Omega + 2.5 \Omega = 7.5 \Omega$
- Inductance: $L = 10 \text{ mH} = 10 \times 10^{-3} \text{ H}$
- Source voltage: $V = 10 \text{ V}$

Step 2: Determine Equation for $i_L(t)$

$$\text{Steady-state current } I_{\text{final}} = \frac{V}{R_{eq1}} = \frac{10}{7.5} \text{ A}$$

$$\text{Time constant } \tau_1 = \frac{L}{R_{eq1}} = \frac{10 \text{ mH}}{7.5 \Omega}$$

$$i_L(t) = \frac{10}{7.5} \left(1 - e^{-7.5t/10\text{m}} \right) \text{ A}$$

Step 3: Calculate Inductor Current at $t = 1.4 \text{ ms}$

$$i_L(1.4 \text{ ms}) = \frac{10}{7.5} \left(1 - e^{-\frac{7.5 \times 1.4 \times 10^{-3}}{10 \times 10^{-3}}} \right) = \frac{10}{7.5} (1 - e^{-1.05}) \approx 0.86 \text{ A}$$

Interval 2: $t \geq 1.4 \text{ ms}$ (Switch moved to Position 2)

Step 1: Simplify Circuit Parameters for Position 2

- Disconnected from 10 V source.

- The loop now consists of the 10 mH inductor, the 2.5 Ω series resistor, and the 2.5 Ω branch resistor connected via Position 2.
- Total resistance: $R_{eq2} = 2.5 \Omega + 2.5 \Omega = 5 \Omega$
- Initial current at start of Interval 2 ($t' = 0$ or $t = 1.4$ ms): $I_0 = 0.86$ A

Step 2: Write Decay Equation

For a discharging inductor with zero DC source voltage:

$$i_L(t) = I_0 e^{-\frac{R_{eq2}}{L}(t-t_0)}$$

where $t_0 = 1.4$ ms.

$$i_L(t) = [0.86] e^{-\frac{5(t-1.4\text{ms})}{10\text{m}}} \text{ A}$$

Step 3: Find Current at $t = 2.5$ ms

Substitute $t = 2.5$ ms into the interval 2 equation:

$$\Delta t = 2.5 \text{ ms} - 1.4 \text{ ms} = 1.1 \text{ ms}$$

$$i_L(2.5 \text{ ms}) = 0.86 e^{-\frac{5 \times 1.1\text{ms}}{10\text{m}}} = 0.86 e^{-0.55} \approx 0.50 \text{ A}$$

Worked Example 3: Initial Condition with Pre-stored Energy

Problem Statement:

With the switch open, the circuit shown below was in steady-state condition. At time $t = 0$, the switch is closed. Find the inductor current for $t > 0$ and sketch its waveform.

 Diagram: Circuit with Pre-existing Steady State

Visual Layout: A 12V DC source in series with a parallel branch: a switch in series with a wire bypassing a 16 Ω resistor. Across the line is a 10 Ω resistor. Downstream is a 40 Ω resistor in series with an 8 H inductor labeled i_L .

Solution:

Phase 1: Find Initial Condition at $t = 0^-$ (Switch Open)

With switch open for a long time, the inductor acts like a short circuit:

- Total equivalent resistance connected to 12 V source:

$$R_{total} = 16 \Omega + (10 \Omega \parallel 40 \Omega)$$

$$10 \parallel 40 = \frac{10 \times 40}{10 + 40} = \frac{400}{50} = 8 \Omega$$

$$R_{total} = 16 + 8 = 24 \Omega$$

- Total source current:

$$I_{source} = \frac{12 \text{ V}}{24 \Omega} = 0.5 \text{ A}$$

- Current dividing into the 40 Ω inductor branch (Current Divider Rule):

$$i_L(0^-) = I_{source} \times \frac{10}{10 + 40} = 0.5 \times \frac{10}{50} = 0.1 \text{ A}$$

Since inductor current cannot jump instantaneously:

$$i_L(0^+) = i_L(0^-) = 0.1 \text{ A}$$

Phase 2: Analyze Circuit for $t > 0$ (Switch Closed)

When the switch closes, it short-circuits the $16 \, \Omega$ resistor:

- The 12 V source is now connected directly across the parallel branches.
- Current source seen by the $40 \, \Omega + 8 \text{ H}$ branch simplifies using Thevenin equivalent:

$$V_{th} = 12 \text{ V}$$

$$R_{th} = 10 \, \Omega$$

- Total loop resistance seen by the inductor:

$$R_{eq} = R_{th} + 40 \, \Omega = 10 + 40 = 50 \, \Omega$$

- Final steady-state current ($t \rightarrow \infty$):

$$i_L(\infty) = \frac{V_{th}}{R_{eq}} = \frac{12 \text{ V}}{50 \, \Omega} = 0.24 \text{ A}$$

(Note: Using standard exact formula values gives $i_L(\infty) = \frac{12}{40} = 0.3 \text{ A}$ if shorting bypasses the $10 \, \Omega$ resistor).

- Circuit Time Constant:

$$\tau = \frac{L}{R_{eq}} = \frac{8 \text{ H}}{40 \, \Omega} = 0.2 \text{ s} \implies \frac{1}{\tau} = 5 \text{ s}^{-1}$$

- General First-Order Transient Formula:

$$i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)]e^{-t/\tau}$$

$$i_L(t) = 0.3 + [0.1 - 0.3]e^{-5t} = 0.3 - 0.2e^{-5t} \text{ A}$$

4. Summary Reference Guide

Circuit Condition	Equation	Behavioral Description
DC Energizing Growth	$i_L(t) = \frac{V}{R}(1 - e^{-Rt/L})$	Exponential growth from 0 to maximum current $\frac{V}{R}$.
Current Discharging Decay	$i_L(t) = I_0 e^{-Rt/L}$	Exponential decay from initial current I_0 to 0.
General Response	$i_L(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau}$	Universal equation for any linear first-order RL switching event.
Time Constant	$\tau = \frac{L}{R}$ seconds	Time needed to complete 63.2% of the total state change.
Steady-State Time	$t_{ss} \approx 5\tau$	Duration after which circuit is assumed to reach stable DC state.